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Doctoral Research Project Oral Presentation

Exploring the Role of Cybersecurity Awareness and Training in Enhancing Sustainable Business Performance in Small and Medium-Sized Enterprises: A Qualitative Case Study

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Background to the Study (1)

SMEs: The Backbone of the Global Economy

- Represent >90% of businesses worldwide
- Drive innovation, development, and job creation
- Key players in supply chains

Digital Transformation – Opportunity & Risk

- Cloud computing & digital platforms boost efficiency & competitiveness
- Increased cyber vulnerability due to limited awareness of threats & defenses

Cybersecurity Threat Landscape

- Only 28% confident in risk management
- 67% targeted by cyberattacks; 58% breached
- ~50% lack clear protection strategies

Background to the Study (2)

Implications for Supply Chains

- SMEs handle sensitive financial/personal data
- Breaches can disrupt entire networks & cause financial loss

Persistent Barriers

- SME Cybersecurity often underestimated due to limited awareness, funds & skilled personnel

Call to Action

- Educate employees, enforce proactive protocols
- Strategic investment in resilience & protection

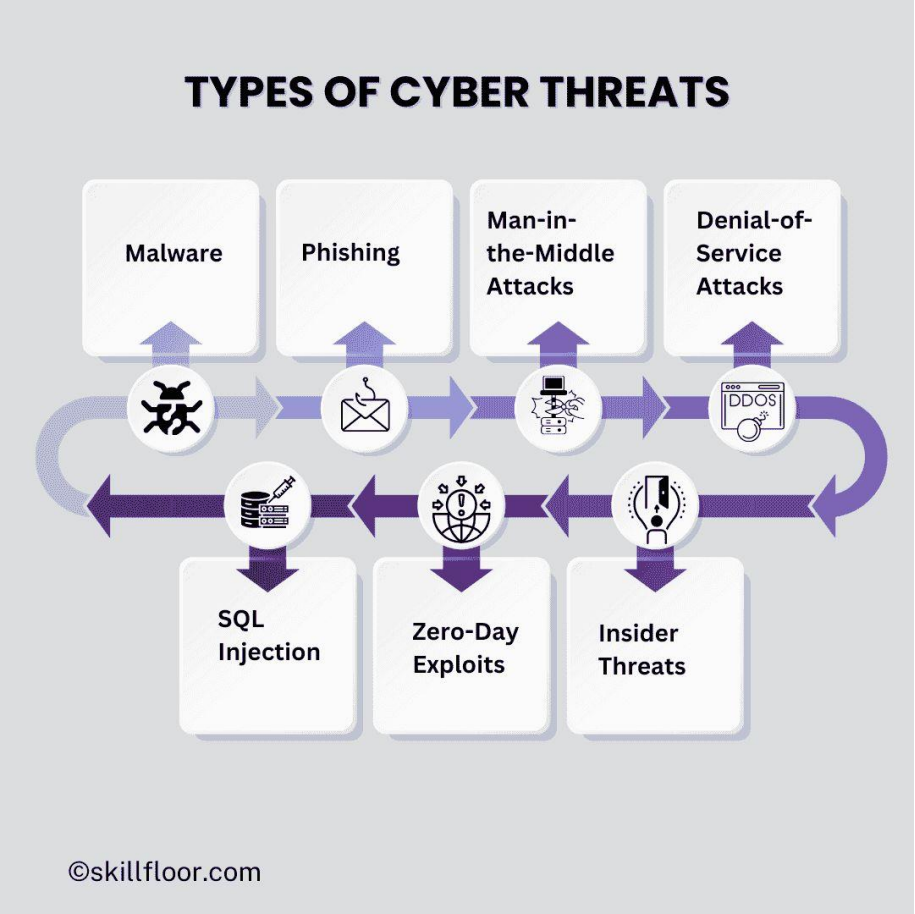
General & Specific Problem Statements

General Problem:

The lack of cybersecurity education and awareness in SMEs heightens vulnerability to cyberattacks capable of disrupting operations and compromising sensitive data.

Specific Problem:

Lack of education and awareness on data security at ***K2 Partnering Solutions*** creating vulnerabilities to cyberattacks, risking sensitive information and potentially disrupting operations.



Purpose of the Study

- Explore how cybersecurity awareness and training impact sustainable business performance at K2.

- Examine the relationship between management and employee awareness of cybersecurity issues and the effectiveness of current training programs.

- Identify how awareness and training initiatives contribute to the company's long-term success and resilience in a digital marketplace.

- Provide insights for strengthening cybersecurity culture to support sustainable business operations.

Significance of the study

Addresses a critical gap in SME cybersecurity by advancing tailored, evidence-based awareness strategies.

Potential to reduce data breaches, enhance customer trust, and foster a robust security culture.

Provides actionable insights for SMEs with limited resources facing heightened cyber threats.

Literature Support (1)

The Disproportionate Cyber Risk Facing SMEs

- **Key Literature Themes:**
  - **Targeted Threats:** SMEs account for **43% of cyberattacks**, underscoring their growing exposure in the digital economy (Verizon, 2023).
  - **Resource Constraints:** Unlike large enterprises, SMEs lack **dedicated IT teams** and **advanced security infrastructures**, leaving them vulnerable (Rawindaran et al., 2023; Faulhaber-Brad Moon, 2023).
  - **Misconceptions & Complacency:** A **false sense of security** among SME owners contributes to underinvestment in cybersecurity (Arroyabe et al., 2024; Rüdell, 2024).
  - **Supply Chain Risk:** SMEs often serve as **entry points to larger organizations**, amplifying systemic risk across industries (Mugwagwa et al., 2024; Sharma, 2025).
- **Scholarly Framing:**

  "The literature consistently identifies SMEs as high-risk entities due to structural vulnerabilities, limited resources, and their strategic position within supply chains." — Tetteh (2024)

Literature Support (2)

- **Key Literature Themes:**
  - **Human Error as a Core Risk:** Poor password hygiene and phishing susceptibility are major breach factors (Javaid et al., 2023; Taherdoost, 2024).
  - **Ineffective Training Models:** Static, one-size-fits-all training fails to drive behavioral change (Nurse, 2021; Abrahams et al., 2024).
  - **Need for Adaptive Learning:** Gamification, simulations, and personalized feedback are underutilized but promising (Taherdoost, 2024; Prümmer et al., 2024).
  - **Leadership Integration Gap:** Despite its importance, leadership involvement in cybersecurity education remains underexplored (Shaw, 2024; Simmons & Sundaresan, 2024).
- **Scholarly Framing:**

  "The literature reveals a critical gap in adaptive, leadership-driven cybersecurity education frameworks tailored to SMEs' unique constraints." — Abrahams et al. (2024).

Research Framework

Integrated TOE and TAM Framework

- Combines the Technology Acceptance Model (TAM) and the Technology-Organization-Environment (TOE)
- Addresses *two core SME challenges*:
  - ✓ Lack of cybersecurity awareness
  - ✓ Limited resources
- **Rationale**
  - ✓ TAM explains **individual adoption behavior**
  - ✓ TOE examines **organizational and environmental influences**
  - ✓ Together → holistic view of cybersecurity adoption in SMEs
- **Expected Outcomes**
  - ✓ Clearer understanding of cybersecurity phenomena in SMEs
  - ✓ Predictive insights for strategic decision-making
  - ✓ Actionable strategies to reduce cyber risks effectively

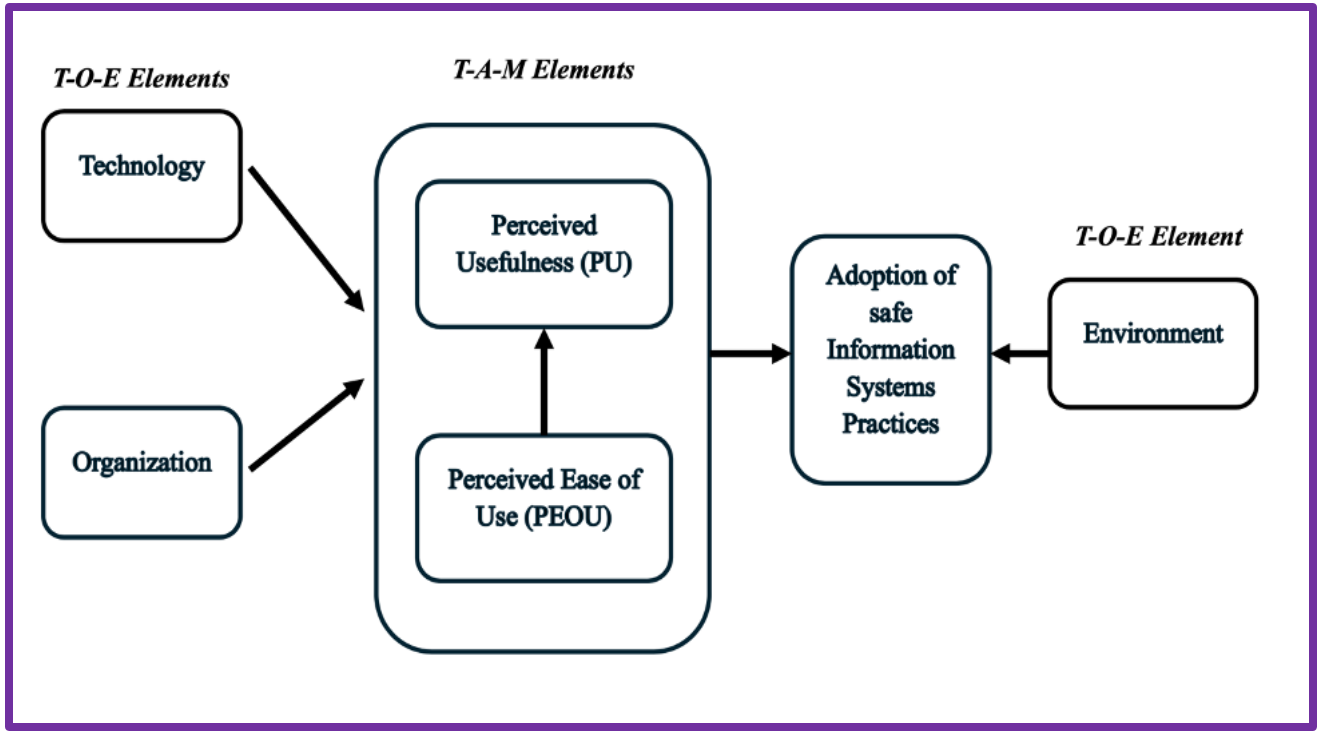


Figure 1. Integrated TAM-TOE Framework
Source: Gangwar, H., Date, H., & Ramaswamy, R. (2015). Understanding determinants of cloud computing adoption using an integrated TAM-TOE model. J. Enterp. Inf. Manag., 28, 107-130.

Assumptions, Limitations and Delimitations

Assumptions	Limitations	Delimitations
• K2 faces notable cyber challenges • Selected participants have relevant knowledge • Documents are accurate, up-to-date & accessible	• Small sample size (10 stakeholders) • Limited access to proprietary data • Interpretive nature of qualitative research leading to interviewer bias	• Focused on K2 only • Internal practices & awareness—no vendor risk analysis • Five-year timeframe for threat relevance • Qualitative scope over quantitative generalization

Research Methodology (1)

Qualitative Approach

- ✓Qualitative methods provide deep insights into organizational behaviors and security practices (Goldberg & Allen, 2015).
- ✓Case study of an SME's cybersecurity posture to uncover vulnerabilities, risk perceptions, and mitigation strategies.
- ✓Focus on the complex interaction of technology, human, and organizational factors.

Purposive Sampling

- ✓Targets key stakeholders: IT personnel, executives, frontline employees.
- ✓Ensures perspectives from those directly involved in cybersecurity decision-making.
- ✓Facilitates rich, multi-dimensional understanding of cybersecurity strategies.

Research Methodology (2)

Mixed-Methods Data Collection

- ✓Combines semi-structured interviews, online surveys, and document analysis.
- ✓Aligns with best practices in cybersecurity research (Yin, 2018).
- ✓Enables a holistic view of cybersecurity dynamics within the SME.

Triangulation for Trustworthiness

- ✓Cross-verifies data from multiple sources to enhance reliability and validity.
- ✓Strengthens confidence in findings through diverse evidence.

Contribution

- ✓Offers actionable insights to improve SME cybersecurity frameworks.
- ✓Bridges academic knowledge with practical security improvements.

Population and Sampling

Study Population: Broad group relevant to cybersecurity in SMEs.

Target Population: Employees at K2 Partnering Solutions, directly impacted by cybersecurity.

Significance:

- Focused on accessible, relevant participants to ensure meaningful insights.
- Aims to inform K2 Partnering Solutions and offer guidance applicable to similar SMEs navigating cybersecurity challenges.

Data Collection and Analysis (1)

Triangulated Data Collection:

- **Semi-structured Interviews**
  - ✓10–15 minutes, flexible format, audio recorded.
  - ✓Purposive sample of 10 participants across departments and hierarchy
- **Online Surveys**
  - ✓Mix of open & closed questions
  - ✓Ensures anonymity, accessibility, and candid responses.
- **Document Analysis**
  - ✓Review of policies, training logs, and incident reports.
  - ✓Provides critical contextual depth.

Data Collection and Analysis (2)

Data Analysis

- **Thematic Analysis Framework**
  - ✓ Braun & Clarke's 6-step process to identify key themes from textual data.
  - ✓ Coding phases: open coding → axial coding → theme development.
  - ✓ Example themes: *Cybersecurity Awareness*, *Training Effectiveness*, *Policy Enforcement*.
- **Analysis Process**
  - ✓ Coding of interviews, surveys, and documents to reveal patterns (e.g., "fragmented cybersecurity culture").
  - ✓ Thematic maps visualize relationships among themes.
  - ✓ Findings linked to research questions and existing literature for rich interpretation.
- **Triangulation for Validity**
  - ✓ Cross-verification of data from interviews, surveys, and documents.
  - ✓ Enhances credibility and reduces bias (Fusch et al., 2018).
  - ✓ Ensures a comprehensive and trustworthy understanding of cybersecurity culture and practices.

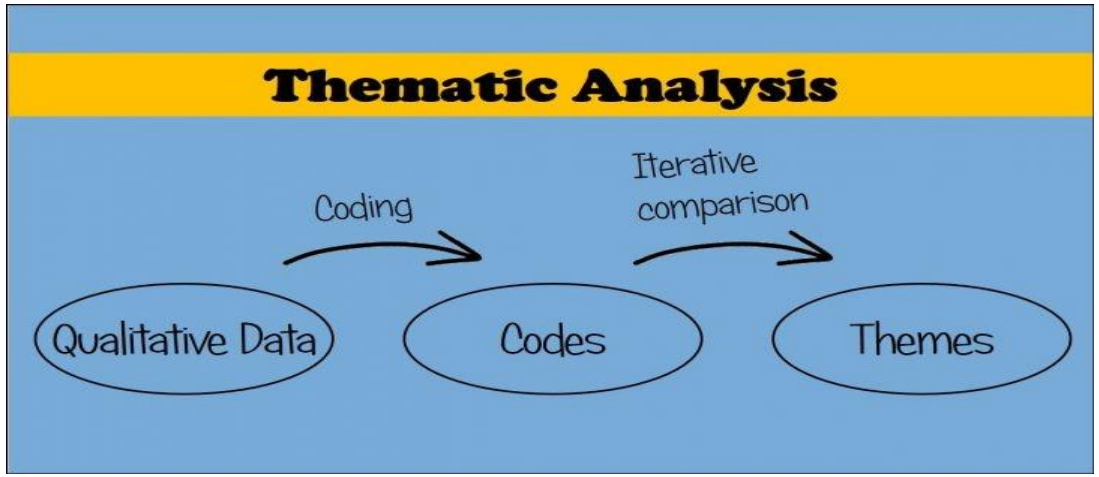
Results, Discussion and Implications (1)

Research Questions

RQ1: What strategies and measures does K2 Partnering Solutions employ to mitigate and counter potential cyber threats to its systems and data?

RQ2: What strategies does K2 Partnering Solutions implement to raise employee awareness and provide training to protect the business from potential cyberattacks?

Thematic Analysis



Four themes emerged:

Organizational Cybersecurity Strategies and Infrastructure	Cybersecurity Awareness and Training	Cybersecurity Culture and Organizational Commitment	Sustainability and Strategic Impact

Results, Discussion & Implications (2)

Theme 1: Organizational Cybersecurity Strategies & Infrastructure

Key Threats:

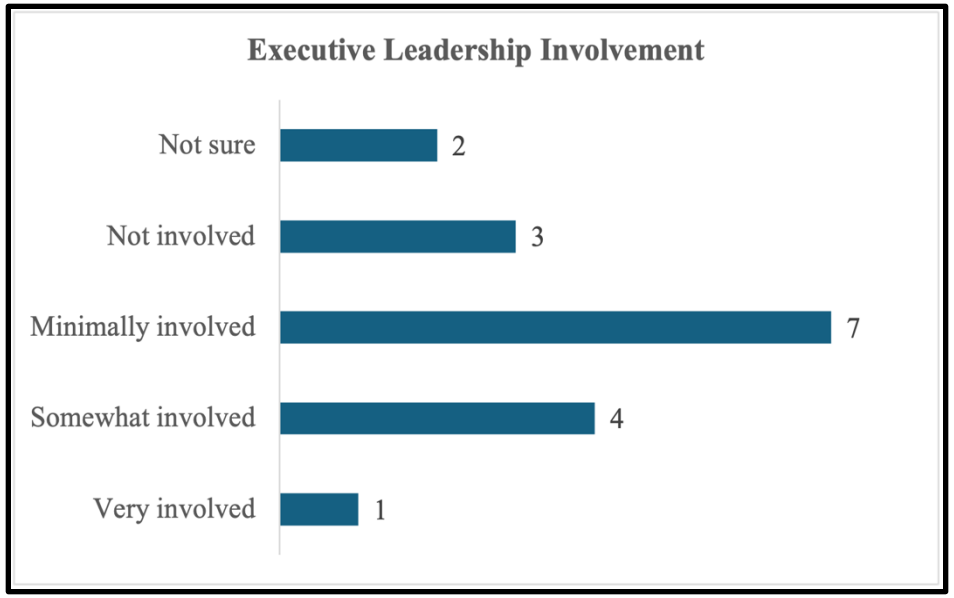
- Phishing & impersonation emails widespread
- Reactive approach via informal alerts (Slack)

Policies & Tech:

- ISO/IEC 27001 framework adopted but poorly communicated
- Basic defenses: spam filters, 2FA; no advanced threat detection

Leadership Involvement:

- Formal policy endorsed, but low visible engagement
- Leadership Involvement perceived as weak across
- Disconnect limits strategic cybersecurity integration



Survey Data

Results, Discussion & Implications (3)

Theme 2: Awareness & Training Culture

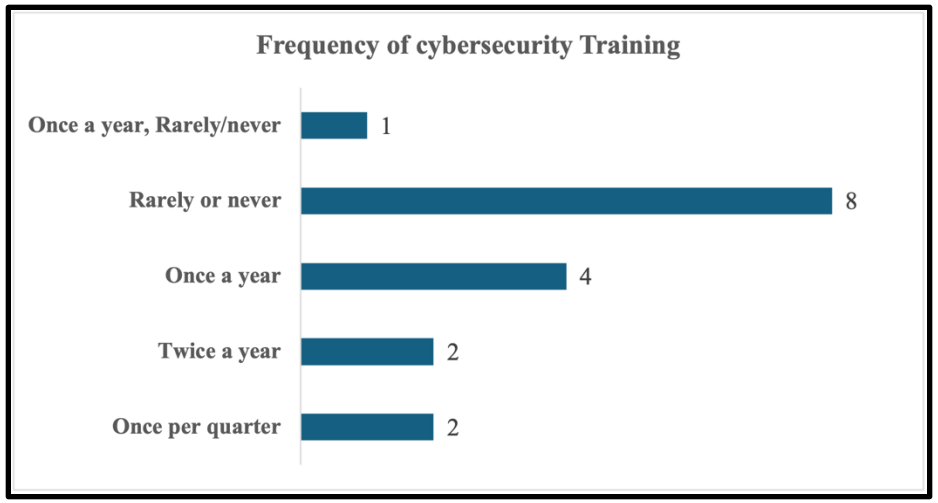
Training Delivery & Effectiveness:

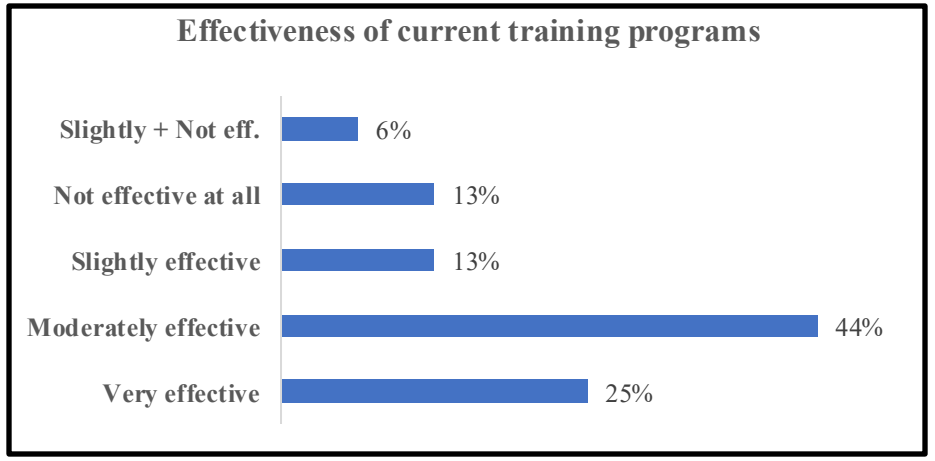
- ✓ Over 50% rarely or never trained; onboarding often the only exposure
- ✓ 44% rated training as only moderately effective

Implications:

- ✓ Implement structured, recurring training programs
- ✓ Use experiential learning (simulations, case studies) over passive formats
- ✓ Introduce evaluation metrics (quizzes, follow-ups) to measure training impact

Survey Data





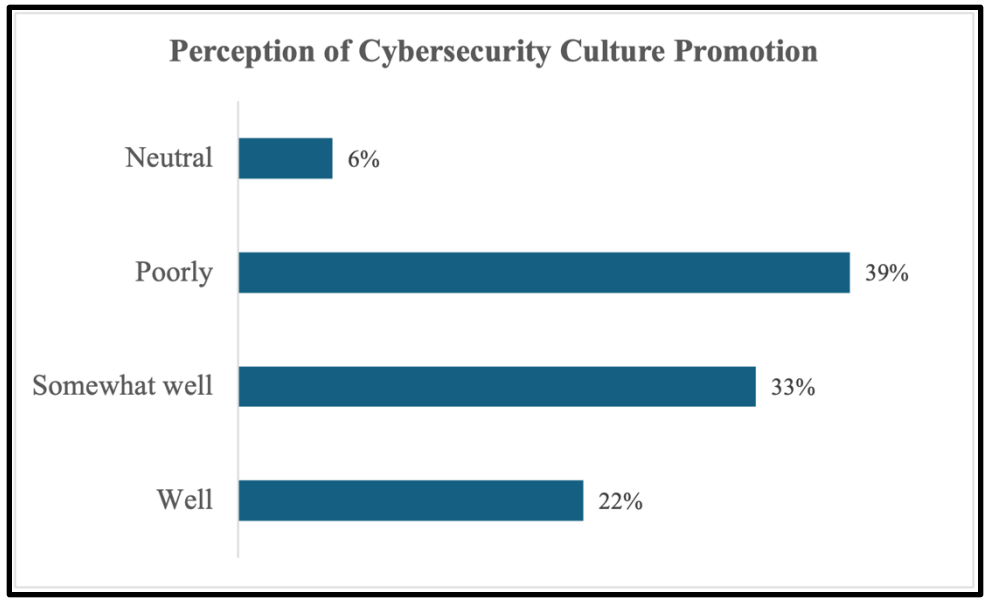
Results, Discussion & Implications (4)

Theme 3: Cybersecurity Culture & Commitment

- ✓Only 22% say cybersecurity culture promoted effectively
- ✓Leadership support procedural, not participatory
- ✓Lack of visible campaigns, drills, peer support

Implications:

- ✓Leadership must role-model secure behaviors
- ✓Launch organization-wide awareness campaigns
- ✓Encourage cross-functional, peer-led security initiatives



Survey Data

Results, Discussion & Implications (5)

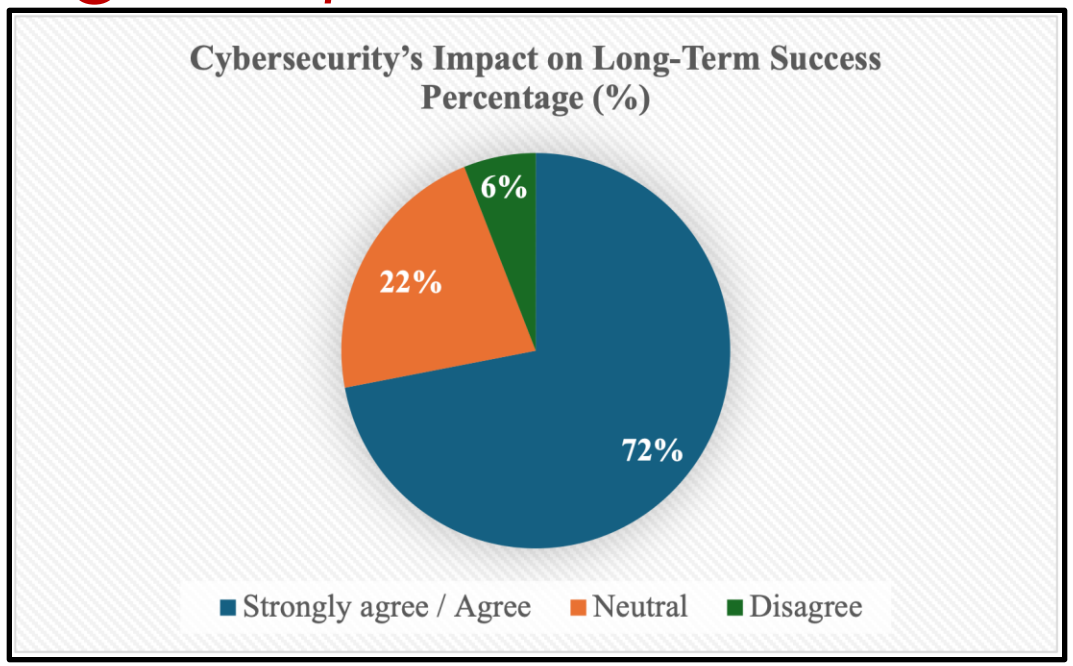
Theme 4: Sustainability & Strategic Impact

Strategic Recognition

- ✓ 72% agree cybersecurity affects long-term success
- ✓ Strategic opportunity: integrate security into business continuity
- ✓ Contractor model makes security critical

Implications:

- ✓ Cybersecurity must shift from reactive to proactive strategic pillar
- ✓ Leadership engagement & culture-building essential for resilience
- ✓ Systematic training, continuous improvement, and compliance audits needed



Recommendations for Future Research

Key Directions for Future Research

Advancing Cybersecurity Research in SMEs

- **Longitudinal & Comparative Studies**
  - Track culture, leadership, and training impact over time.
  - Compare SMEs across industries/regions to identify context-specific maturity drivers.
- **Socio-Technical & Leadership Dynamics**
  - Examine informal communication (e.g., Slack) as agile awareness channels.
  - Investigate executive decision-making, risk perception, and tech adoption barriers.
- **Innovative Training & Governance Models**
  - Test gamified, scenario-based training for behavioral impact.
  - Integrate cybersecurity into enterprise risk management and cultural norms.

Closing Remarks

01

Cybersecurity is integral to SME sustainability in the digital economy.

02

Education and awareness serve as the most cost-effective and impactful defenses.

03

This study contributes actionable, context-specific strategies to strengthen SME resilience.

04

A proactive cybersecurity culture safeguards operational continuity and stakeholder trust.

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Q & As